Sustainable Cloud Gaming Infrastructure: A Review of Renewable Energy Integration, Optimization Frameworks, and Computing-Power Synergy

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Abstract

The convergence of cloud gaming, artificial intelligence-driven game development, and global carbon neutrality imperatives has positioned game data centers at the epicenter of a critical energy transition challenge. This review provides a comprehensive and systematic examination of renewable energy integration for game data centers, with particular emphasis on the emerging paradigm of computing-power synergy — the coordinated management of computational workloads and green energy supply. We begin by characterizing the unique energy consumption patterns, carbon footprints, and operational constraints of game data centers, distinguishing them from general-purpose cloud infrastructure through analysis of latency sensitivity, workload burstiness, and GPU intensity. We then survey the theoretical foundations of optimization under renewable uncertainty, including robust optimization, distributionally robust optimization, decision-dependent uncertainty frameworks, and stochastic programming approaches that have been developed in the power systems literature. The review proceeds to examine game-theoretic formulations for multi-actor resource allocation, covering non-cooperative games, cooperative games, Stackelberg games, negotiation games, and asymmetric Nash bargaining solutions. We synthesize recent advances in prediction-dispatch integration, including machine learning-based forecasting, reinforcement learning for real-time scheduling, and unified optimization frameworks. A comprehensive survey of enabling technologies covers IT equipment flexibility through GPU sharing and containerization, infrastructure flexibility through energy storage and cooling systems, waste heat recovery through district heating integration and organic Rankine cycles, and energy-efficient hardware architectures. We critically review representative engineering deployments across China, Europe, Japan, and the United States, including the 100% renewable data centers in Qinghai, the Ulaanqab green power direct-supply project, the East Data West Computing initiative, the DATAZERO project, Ubitus nuclear-powered cloud gaming infrastructure, Boosteroid's renewable-powered Polish deployment, and the Hainan International Data Center. We identify persistent challenges in quality-of-service preservation under renewable constraints, cross-domain coordination across electricity, carbon, and green certificate markets, scalability of optimization methods, and the tension between workload migration and latency bounds. Finally, we outline future research directions including digital twin-enabled energy management, net-negative carbon gaming ecosystems, AI-native optimization architectures, sustainable game design integration, and edge-cloud continuum optimization. This review bridges the gap between theoretical advances in power system optimization and engineering practice in sustainable gaming infrastructure, providing a coherent framework for the transition from green electricity procurement to truly green computing.

Keywords: Game data centers, Cloud gaming, Renewable energy integration, Computing-power synergy, Robust optimization, Game theory

1 Introduction

1.1 Background and Motivation

The global gaming industry has undergone a profound transformation over the past decade, driven by the proliferation of cloud gaming services, the rise of generative artificial intelligence for content creation, and the increasing computational demands of realistic game physics, graphics rendering, and immersive virtual worlds (1; 2). Cloud gaming platforms such as Boosteroid, NVIDIA GeForce NOW, Xbox Cloud Gaming, and major cloud providers including Alibaba Cloud and Amazon Web Services now deliver high-fidelity gaming experiences directly to end-users without requiring expensive local hardware (3). This architectural shift transfers the computational burden from consumer devices to geographically distributed data centers, fundamentally reshaping both the economics and the environmental footprint of the gaming industry (4).

The shift toward cloud-based gaming infrastructure represents a paradigm change with profound energy implications. Traditional gaming relied on locally installed software running on consumer hardware, where energy consumption was distributed across millions of devices. Cloud gaming centralizes computation in large-scale data centers, achieving economies of scale in hardware utilization but concentrating energy demand in facilities that require substantial power infrastructure and cooling systems (5). The computational intensity of modern game workloads — particularly those involving real-time ray tracing, physics simulation, and AI-driven non-player character behavior — demands high-performance GPU resources, which consume significantly more power per compute unit than conventional CPU-based cloud workloads (6; 7).

Concurrently, the global push for carbon neutrality has accelerated dramatically. The Paris Agreement's targets, national net-zero commitments, and growing investor and consumer pressure on environmental, social, and governance (ESG) performance have compelled industries across all sectors to decarbonize their operations. The information and communication technology sector, which accounts for an estimated 2-4% of global greenhouse gas emissions, faces particular scrutiny due to its rapid growth trajectory (8). Data centers, as the physical infrastructure underpinning digital services, represent the largest and fastest-growing component of the ICT sector's energy consumption, with China alone consuming 199.07 TWh in 2020 projected to reach 490.18 TWh by 2030 (8).

1.2 The Game Data Center Challenge

The intersection of cloud gaming growth and decarbonization imperatives creates a unique and challenging problem: how to power latency-sensitive, GPU-intensive, and bursty game workloads with inherently intermittent and variable renewable energy sources such as wind and solar (9; 10). Unlike batch processing workloads that can be deferred to periods of high renewable generation, or analytics workloads that can be migrated across time zones to follow the sun, game workloads must be served in real time to maintain acceptable quality of experience (11; 12).

Several distinguishing characteristics of game data centers complicate renewable energy integration. First, the latency sensitivity of cloud gaming imposes strict geographic constraints on workload placement. A significant fraction of the population is not within 80 ms round-trip time of a typical data center, and edge-assisted deployment architectures are necessary to maintain playability (12; 13). This geographic constraint limits the ability to migrate workloads to regions with abundant renewable generation, a strategy commonly employed for less latency-sensitive workloads. Second, game workloads exhibit high burstiness and unpredictability, driven by player behavior patterns, game launch events, and the diurnal cycle of gaming activity (4). This variability complicates both load forecasting and renewable generation matching. Third, the computational intensity of modern game rendering and AI inference relies heavily on GPU resources, which have different power consumption profiles and thermal characteristics than CPU-based workloads (6; 14).

1.3 The Computing-Power Synergy Paradigm

The central thesis of this review is that addressing the game data center energy challenge requires a paradigm shift from passive renewable energy procurement to active coordination of computational resources with renewable energy availability. We term this paradigm computing-power synergy, which encompasses the integrated management of computational workloads, power consumption, and carbon emissions across the data center lifecycle (15; 16).

The computing-power synergy paradigm recognizes that data centers are not simply passive loads on the electrical grid but rather flexible resources that can modulate their energy consumption through multiple mechanisms. At the workload level, computational tasks can be scheduled, migrated, or provisioned in response to renewable availability (17; 18). At the infrastructure level, energy storage, cooling systems, and backup power sources can be orchestrated to buffer renewable variability (19; 20). At the grid interaction level, data centers can participate in demand response programs, frequency regulation markets, and ancillary services (16; 21). This multi-level flexibility creates a rich design space for optimizing the balance between computational performance, energy cost, and carbon emissions.

The computing-power synergy paradigm has gained substantial momentum through both academic research and engineering practice. The Chinese government's East Data West Computing national initiative explicitly aims to locate computational infrastructure in western regions with abundant renewable resources, connecting them with eastern demand centers through high-bandwidth networks (22; 23). The Qinghai green data center, China's first 100% renewable energy traceable zero-carbon data center, has completed the industry's first cross-region computing-power collaborative scheduling commercial trial, demonstrating the practical feasibility of the paradigm (24). These developments signal a fundamental transformation in how data center energy is conceived and managed.

1.4 Scope and Contributions

This review provides a comprehensive and systematic examination of renewable energy integration for game data centers, with particular emphasis on the computing-power synergy paradigm. Unlike prior surveys that treat data centers generically (9; 10; 25; 26; 27; 28), we explicitly address the unique characteristics of game workloads and their implications for energy management. Our specific contributions are fourfold.

First, we provide a detailed characterization of game data center energy demand, distinguishing cloud gaming from other data center workload types through analysis of latency sensitivity, load predictability, migration flexibility, and computational intensity. This characterization establishes the foundation for understanding why game data centers present distinct challenges for renewable integration.

Second, we offer a structured survey of mathematical optimization frameworks for renewable-powered data centers, organized by their treatment of uncertainty and coordination structure. We cover robust optimization, distributionally robust optimization, decision-dependent uncertainty, stochastic programming, Stackelberg games, cooperative and non-cooperative games, negotiation games, and distributed collaborative optimization. This survey bridges the power systems optimization literature and the data center resource management literature.

Third, we synthesize recent advances in enabling technologies, including prediction-dispatch integration, flexibility extraction from IT equipment and infrastructure, waste heat recovery, and energy-efficient hardware. We provide critical analysis of engineering deployments across China, Europe, Japan, and the United States, grounding theoretical advances in practical implementation.

Fourth, we identify persistent challenges and outline future research directions, including digital twin-enabled energy management, net-negative carbon gaming ecosystems, AI-native optimization architectures, sustainable game design integration, and edge-cloud continuum optimization. This forward-looking perspective aims to guide research and investment in sustainable gaming infrastructure.

1.5 Organization

The remainder of this paper is organized as follows. Section 2 characterizes game data center energy consumption patterns, workload characteristics, and carbon footprints, distinguishing game workloads from other data center applications. Section 3 provides a comprehensive survey of mathematical optimization frameworks for renewable-powered data centers, covering uncertainty handling, game-theoretic approaches, and prediction-dispatch integration. Section 5 surveys enabling technologies and engineering practices, including flexibility extraction, waste heat recovery, hardware innovations, and representative deployments. Section 6 identifies key challenges and future research directions. Section 7 concludes the paper.

2 Characterization of Game Data Center Energy Demand

Understanding the energy consumption patterns, operational constraints, and environmental footprint of game data centers is essential for designing effective renewable energy integration strategies. This section provides a comprehensive characterization of these dimensions, distinguishing game workloads from general-purpose cloud computing and establishing the foundation for the optimization frameworks surveyed in subsequent sections.

2.1 Cloud Gaming Architecture and Infrastructure

Cloud gaming systems consist of geographically distributed data centers that render game frames, encode video streams, and transmit them to end-user devices with minimal latency (1; 3). The cloud gaming architecture differs from traditional content delivery networks and web services in several important respects. The end-to-end pipeline encompasses game state processing, graphics rendering, video encoding, network transmission, and client-side decoding, all of which must complete within the perceptual latency budget of approximately 80-150 ms for acceptable quality of experience (12).

The computational demands of cloud gaming have evolved substantially with the increasing graphical fidelity of modern games. Real-time ray tracing, which simulates the physical behavior of light for photorealistic rendering, requires orders of magnitude more computational power than traditional rasterization techniques. AI-driven non-player character behavior, procedural content generation, and real-time physics simulation further increase the computational load (7; 13). These workloads are predominantly executed on GPUs, which have significantly higher thermal design power and power consumption profiles than CPUs. The energy intensity of GPU-based computation has driven substantial innovation in resource sharing and isolation mechanisms.

The cloud gaming infrastructure also differs in its distribution pattern. Unlike content delivery networks that can cache static content at the edge, cloud gaming requires real-time computation at the edge or near-edge to maintain acceptable latency. The Edge Cloud hybrid architecture (12) demonstrates that a significant fraction of the population is not within 80 ms round-trip time of a typical data center, motivating the deployment of edge-servers that complement central data centers. This distributed infrastructure creates both challenges and opportunities for renewable energy integration. On one hand, the geographic dispersion of computational resources complicates centralized energy management. On the other hand, the presence of multiple locations provides opportunities for workload migration and renewable portfolio diversification (4; 17).

2.2 Workload Characteristics of Cloud Gaming

The workload characteristics of cloud gaming services exhibit distinctive patterns that fundamentally shape their energy management requirements. We analyze these characteristics along several key dimensions.

Table 1 compares cloud gaming workloads with other common data center applications across seven dimensions. The comparison reveals that cloud gaming exhibits the most restrictive combination of characteristics for renewable energy integration: it is highly latency-sensitive, poorly predictable, geographically constrained, and computationally

Table 1: Comparison of Workload Characteristics Across Data Center Types

Characteristic	Cloud Gaming	Batch Processing	Web Services	AI Training
Latency requirement (ms)	< 80	$10^3 - 10^6$	< 200	< 1000
Load predictability	Low	High	Medium	Medium
Migration flexibility	Very Low	High	Medium	Low
GPU intensity	Very High	Low	Low	Very High
Interactivity	High	None	Low	None
Burstiness	High	Low	Medium	Low
Edge dependency	High	Low	Medium	Low

intensive. These characteristics imply that conventional renewable integration strategies that rely on workload deferral or long-distance migration are less applicable to cloud gaming.

The latency sensitivity dimension deserves particular attention. Cloud gaming requires consistent sub-80 ms round-trip latency for acceptable quality of experience (2; 12). The SARA-SDN framework (11) demonstrates that game-state aware resource allocation can improve average Quality of Experience by 1.26 points on a 0–5 scale through dynamic consideration of delay, bandwidth, and game state. This latency sensitivity imposes fundamental constraints on workload migration strategies that might otherwise enable load following to renewable generation.

2.3 Energy Consumption Patterns

The energy consumption of game data centers exhibits several distinctive patterns that inform renewable integration strategies. Understanding these patterns requires analysis at multiple temporal scales and system levels.

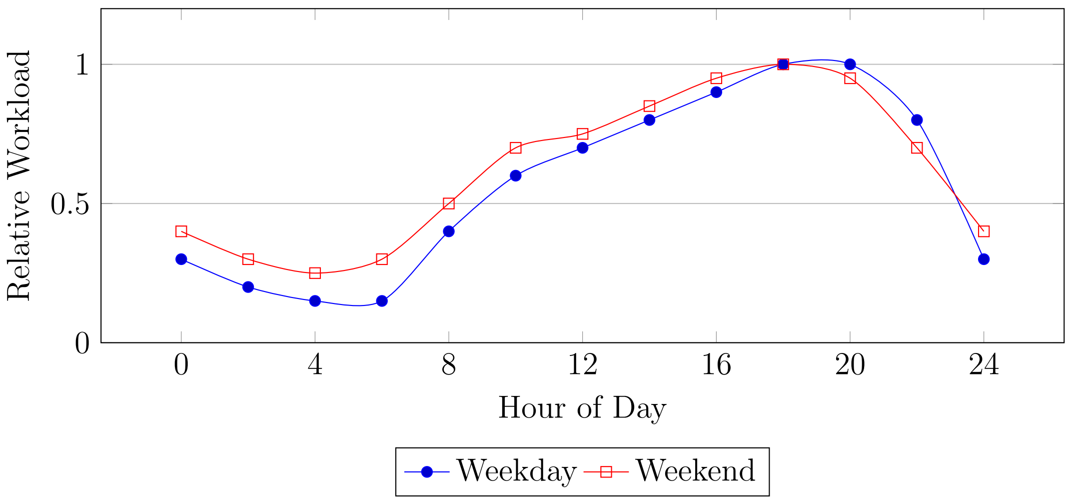

Figure 1: Diurnal workload patterns for cloud gaming services showing weekday and weekend variations.

At the diurnal scale, cloud gaming workloads exhibit pronounced daily cycles driven by player behavior. Figure 1 illustrates typical weekday and weekend patterns, with peak usage occurring during evening hours and weekends showing higher usage during daytime periods. This diurnal pattern has important implications for renewable integration: solar generation peaks during midday when gaming demand is moderate, while wind generation may peak during evening hours in some regions, creating opportunities for temporal alignment.

The burstiness of gaming workloads at shorter timescales presents additional challenges. Game launch events, esports tournaments, and viral content can create sudden demand spikes that exceed normal capacity planning margins (4). The ERPOL approach for edge resource provisioning in cloud gaming (5) addresses this challenge through online optimization that does not require future workload information, achieving significant cost savings while managing the trade-off between provisioning cost and queuing issues.

2.4 Carbon Footprint and Environmental Impact

The carbon footprint of game data centers encompasses operational emissions from electricity consumption, embodied emissions from hardware manufacturing, and emissions from supporting infrastructure such as cooling systems. The rapid growth in data center electricity consumption has made operational emissions the dominant component, with lifecycle assessments indicating that operational emissions account for approximately 70-80% of total lifecycle emissions over a typical data center lifespan.

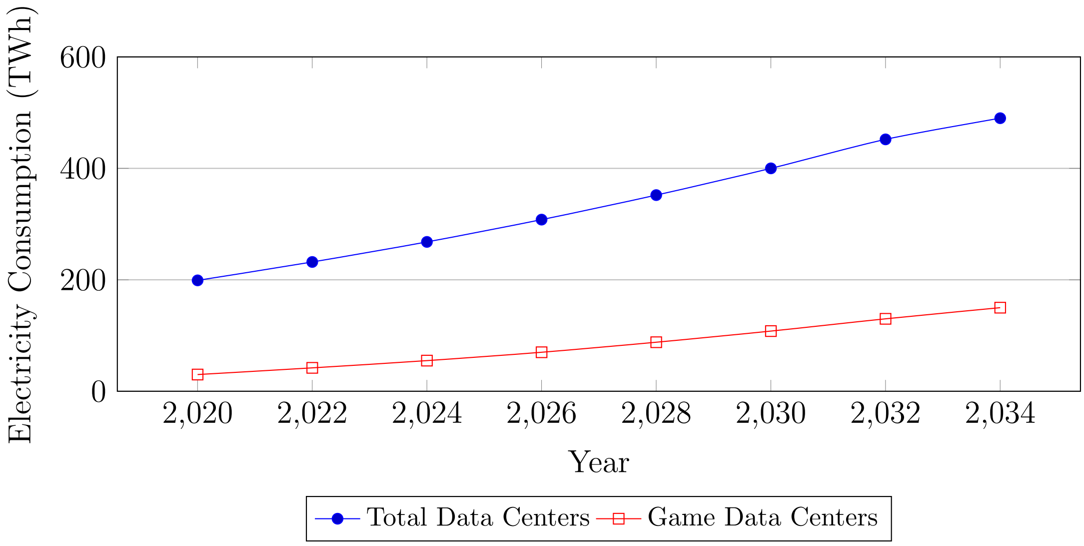

Figure 2: Projected electricity consumption of data centers in China with game data center estimates (8).

Figure 2 illustrates the projected growth trajectory of data center electricity consumption in China, with game data center consumption estimated based on the gaming sector's share of digital infrastructure. The projected growth from 199.07 TWh in 2020 to 490.18 TWh by 2030 represents a compound annual growth rate of approximately 9.4%. Game data centers are projected to account for an increasing share of this consumption as cloud gaming penetration grows and game production becomes more computationally intensive.

The carbon footprint implications are substantial. Assuming current grid emission factors of approximately 0.5 kg CO_2 per kWh, the projected 490.18 TWh would translate to approximately 245 million tonnes of CO_2 annually by 2030. This represents approximately 2.5% of China's total CO_2 emissions, underscoring the urgency of decarbonizing digital infrastructure. The carbon footprint is further amplified by the embodied emissions of the specialized hardware required for game workloads, including GPUs, high-bandwidth memory, and specialized networking equipment.

Table 2: Energy and Carbon Metrics for Representative Game Data Center Workloads

Workload Type	**Avg. Power (kW)**	**PUE**	**Carbon Intensity (kg CO2/MWh)**
Cloud Gaming Render	0.8–1.2	1.3–1.6	450–650
Game AI Training	1.0–1.5	1.4–1.7	500–700
Game Asset Processing	0.5–0.8	1.2–1.5	400–600
Traditional Web Gaming	0.3–0.5	1.2–1.4	350–500

Table 2 presents energy and carbon metrics for representative game data center workloads, highlighting the substantial variation in energy intensity across different types of game-related computation. Cloud gaming render workloads, which require real-time GPU execution, exhibit the highest power consumption and carbon intensity per server, while traditional web gaming has significantly lower requirements.

2.5 Comparison with General-Purpose Data Centers

Game data centers differ from general-purpose data centers in ways that fundamentally affect renewable integration strategies. Table 1 provides a comprehensive comparison across multiple dimensions. The most significant differences include:

Geographic Distribution. General-purpose cloud workloads can be distributed globally with relatively little impact on user experience, enabling workload migration to follow renewable generation. Game workloads are geographically constrained by latency requirements, limiting the scope for migration-based renewable integration (4; 17).

Time Sensitivity. General-purpose workloads often have temporal flexibility that enables deferral to periods of high renewable generation. Game workloads are time-sensitive and cannot be significantly deferred without affecting user experience.

Computational Intensity. Game workloads consume more power per computation than general-purpose workloads due to GPU utilization (6; 14). This high power density creates cooling challenges and increases the economic value of energy efficiency improvements.

Load Predictability. General-purpose workloads often exhibit predictable patterns that facilitate renewable integration planning. Game workloads are subject to unpredictable demand spikes and rapid changes in player behavior.

These differences imply that renewable integration strategies developed for general-purpose data centers cannot be directly applied to game data centers. Instead, game data centers require specialized approaches that account for their unique characteristics, including edge computing architectures, latency-aware migration policies, and GPU-aware power management.

3 Mathematical Optimization for Renewable-Powered Data Centers

The integration of renewable energy into data center power systems requires sophisticated optimization frameworks that can handle uncertainty, multi-timescale dynamics, and multi-actor interactions. This section provides a comprehensive survey of the key mathematical approaches, organized by their treatment of uncertainty, coordination structure, and application context.

3.1 Handling Renewable Uncertainty

The fundamental challenge in renewable-powered data centers is the inherent uncertainty of wind and solar generation. Unlike conventional power sources, renewables cannot be dispatched deterministically, necessitating optimization methods that explicitly account for uncertainty. This subsection reviews the major approaches for uncertainty handling in power system and data center optimization.

3.1.1 Robust Optimization

Robust optimization addresses uncertainty by seeking solutions that remain feasible for all realizations within a specified uncertainty set. In the context of power dispatch, this approach has been extensively developed for wind-integrated systems. Qu et al. (29) propose distributionally robust energy and reserve dispatch frameworks that integrate distributed predictions of renewable energy, providing guarantees under ambiguity about the true probability distribution. Their approximately adaptive approach (30) enables two-stage decisions where reserve procurement can adapt to observed renewable generation.

The segmented distributionally robust optimization method for real-time power dispatch with wind uncertainty (31) achieves improved computational efficiency while maintaining robustness to wind uncertainty. The key innovation is the segmentation of the uncertainty set based on prediction error characteristics, enabling more efficient solution of the robust optimization problem without sacrificing robustness guarantees. This method has direct applicability to data center energy management, where day-ahead and real-time dispatch decisions must account for renewable generation uncertainty.

The environmental-economic unit commitment framework (32) extends robust optimization to co-manage pollution diffusion constraints, demonstrating the applicability to multi-objective settings where both economic and environmental objectives must be satisfied simultaneously. The stochastic robust real-time power dispatch framework (33) combines stochastic programming and robust optimization using difference-of-convexity optimization, achieving efficient handling of wind power uncertainty in real-time dispatch. This hybrid approach is particularly relevant for data centers where both the expected cost and worst-case robustness matter.

The review by Zhong et al. (34) provides a comprehensive survey of efficient optimization methods for high-renewable power system operation, covering solver tuning, heuristic algorithms, data-driven techniques, cutting plane methods, Benders decomposition, and Lagrangian relaxation. The review identifies the integration of physical mechanisms with data-driven approaches as the key future direction, a conclusion that applies equally to data center energy management.

3.1.2 Distributionally Robust Optimization

Distributionally Robust Optimization (DRO) bridges the gap between stochastic programming (which requires full distributional knowledge) and robust optimization (which ignores distributional information beyond support). DRO constructs ambiguity sets around predicted distributions, optimizing for the worst-case distribution within the ambiguity set. This approach achieves a balance between conservatism and performance, making it particularly well-suited for data center energy management where both the renewable generation distribution and the workload distribution are uncertain.

The distributionally robust energy and reserve dispatch framework (29) constructs ambiguity sets using predicted distributions of renewable generation. The approximately adaptive DRO approach (30) further enables two-stage decisions where reserve procurement can adapt to observed renewable generation, providing more flexible and cost-effective operation. The segmented DRO method (31) partitions the uncertainty space to improve computational tractability while maintaining robustness guarantees.

3.1.3 Decision-Dependent Uncertainty

Recent work by Zhang et al. (35) highlights an important nuance: renewable uncertainty can be decision-dependent. The concept of separability bridges decision-dependent uncertainties (DDUs) with conventional decision-independent uncertainties (DIUs), enabling existing algorithms to address DDUs. This has significant implications for data center dispatch, where the decision to migrate workloads may itself influence the renewable generation available at destination sites due to curtailment, transmission constraints, or market interactions.

The decision-dependent uncertainty framework in power systems with high-share renewables (35) demonstrates that the dispatch decision can affect the uncertainty set of renewable generation through both physical and market mechanisms. Applications in wind power and demand response demonstrate significant frequency regulation and robustness improvements, suggesting that accounting for decision-dependence can improve data center energy management outcomes.

3.1.4 Stochastic Programming and Multi-Stage Optimization

Stochastic programming explicitly models uncertainty through scenarios, optimizing the expected value of the objective function. The stochastic robust real-time power dispatch framework (33) combines stochastic programming and robust optimization, using difference-of-convexity optimization to handle wind power uncertainty efficiently. This approach is computationally tractable for real-time applications and provides solutions that balance expected performance against worst-case robustness.

The multi-stage nature of energy management decisions — day-ahead scheduling, hour-ahead adjustments, and real-time dispatch — naturally calls for multi-stage stochastic programming or robust optimization formulations. The approximately adaptive DRO approach (30) provides a practical solution by enabling adaptation at the second stage, which is suitable for data center applications where day-ahead workload planning and real-time energy adjustments are made at different timescales.

3.2 Data Center and Grid Coordination Frameworks

The interaction between data centers and power grids creates a multi-agent system requiring coordination mechanisms that respect both computational service level agreements and grid operational constraints. This subsection surveys coordination frameworks that have been developed for renewable-powered data centers.

3.2.1 Stackelberg Game Formulations

The joint energy and computation workload management framework (15) formulates a three-layer Stackelberg game modeling interactions among energy markets, data center operators, and smart grids. This structure captures the hierarchical decision-making where the grid sets prices, data centers respond with workload allocations, and end-users adapt their demand. The solution yields dynamic workload allocation strategies that minimize both energy consumption and carbon emission costs.

The Stackelberg game formulation is particularly appropriate for data center energy management because it captures the asymmetric power relationship between the grid (which sets prices and operating constraints) and data center operators (who are price-takers in the energy market but have control over their computational resources). The game-theoretic solution provides insights into the equilibrium outcomes and the efficiency of market-based coordination.

3.2.2 Bi-Level Optimization and ADMM

The spatial-temporal flexibility-enhanced demand response method for Internet data centers under coupled power, green certificate, and carbon markets (16) formulates a bi-level optimization problem. The upper level addresses market decisions, while the lower level addresses data center operations. A refined workload allocation model captures workload heterogeneity and hybrid cooling systems. The bi-level game strategy, solved using an improved alternating direction method of multipliers (ADMM), achieves 10.5% cost reduction and 10.2% carbon emission reduction in case studies.

The ADMM solution approach is particularly appealing for data center applications because it preserves the privacy of individual agents (data centers can optimize their own operations without revealing sensitive workload information), enables distributed computation, and has convergence guarantees under mild conditions. The enhanced self-adaptive PCB-ADMM algorithm used by Yang et al. (21) further improves convergence speed and solution quality.

3.2.3 Distributed Collaborative Optimization

The review by Ma et al. (28) surveys multi-agent system-based distributed collaborative optimization methods for economic dispatch in new power systems. Covering optimization algorithms under undirected/directed graphs, communication constraints, cybersecurity concerns, and heterogeneous objectives, this body of work provides the theoretical foundation for coordinating geographically distributed data centers as autonomous agents.

Table 3 compares the major coordination frameworks along three dimensions: decision level, privacy preservation, and scalability. The comparison reveals trade-offs that inform the choice of framework for specific applications. Distributed optimization approaches

Table 3: Comparison of Coordination Frameworks for Data Center-Grid Interaction

Framework	Decision Level	Privacy Preserving	Scalability
Stackelberg Game	Hierarchical	Medium	High
Bi-Level ADMM	Two-Stage	High	Medium
Distributed Consensus	Peer-to-Peer	High	High
Centralized Optimization	Single-Level	Low	Low

offer superior privacy preservation and scalability but may have slower convergence and less certain optimality guarantees.

3.3 Game-Theoretic Resource Allocation

Game theory provides a natural framework for analyzing resource allocation among self-interested actors, a scenario that arises in multi-tenant data centers and competitive cloud gaming markets. This subsection reviews game-theoretic approaches for resource allocation in renewable-powered data centers.

3.3.1 Non-Cooperative Games

The game-based traffic exchange framework (36) proposes non-cooperative game models for homogeneous data centers, enabling workload exchange according to green energy availability. In the non-cooperative setting, each data center independently decides its workload allocation to minimize its own cost, leading to a Nash equilibrium. Experimental results demonstrate reduction in energy costs and improvement in average utility through this decentralized mechanism.

Non-cooperative game formulations are appropriate for competitive cloud gaming markets where each provider independently optimizes its own operations. The Nash equilibrium solution provides insights into the stability and efficiency of market outcomes. However, non-cooperative games may lead to suboptimal collective outcomes due to the tragedy of the commons, motivating the need for cooperative game formulations.

3.3.2 Cooperative Games and Coalition Formation

The game-based traffic exchange framework (36) also proposes cooperative game models for heterogeneous data centers, where data centers can form coalitions to share renewable energy resources. Cooperative game theory provides mechanisms for allocating the gains from cooperation among coalition members, ensuring that all participants benefit from collaboration.

The cooperative game formulation is particularly relevant for data centers within the same corporate group or for data centers participating in renewable energy purchasing cooperatives. The coalition formation process enables data centers with complementary renewable generation profiles to achieve higher renewable penetration than would be possible individually.

3.3.3 Negotiation Games

The negotiation game framework for joint IT and energy management in green datacenters (19; 37) enables data centers to negotiate power allocation with smart grids based on multiple metrics including green energy consumption, critical applications, and energy storage state. This approach allows decentralized decision-making while preserving privacy and enabling multiple stakeholders to express their preferences.

The negotiation game approach (19) is particularly appropriate for scenarios where data centers and grid operators have different objectives and information. The negotiation process enables the discovery of mutually beneficial agreements without requiring complete information sharing. The DATAZERO project (20) implements this approach as part of a comprehensive framework for zero-emission data centers using renewable energy.

3.3.4 Asymmetric Nash Bargaining

The cooperative framework for interconnected data center prosumers with PV and battery storage (21) formulates the problem as an asymmetric Nash bargaining cooperative game maximizing overall payoff. The asymmetric Nash bargaining solution captures differences in bargaining power among data centers, reflecting real-world heterogeneity in market position and resource endowments. An enhanced self-adaptive PCB-ADMM algorithm preserves privacy while solving the bargaining problem, achieving 6.63% reduction in overall cost in case studies.

The asymmetric Nash bargaining solution has several appealing properties for data center resource allocation: it is Pareto optimal, satisfies symmetry, is scale invariant, and is independent of irrelevant alternatives. These properties ensure that the bargaining solution is fair and efficient.

3.3.5 Computing Power Network Scheduling

The resource scheduling model in computing power network using game theory (18) addresses conflicting demands among data processing tasks, computing resources, clean electricity, and network resources. The model decreases thermal electricity utilization and enhances task completion quality through game-theoretic allocation of computing tasks to clean electricity-rich western data centers in China.

This approach is directly relevant to the East Data West Computing initiative, where computational tasks from eastern demand centers are scheduled to western data centers with abundant renewable resources. The game-theoretic formulation ensures that allocation decisions are rational and stable, accounting for the diverse objectives of different stakeholders.

Figure 3 provides a qualitative comparison of the computational complexity of different optimization frameworks, highlighting the trade-offs between modeling sophistication and computational tractability.

3.4 Prediction-Dispatch Integration

The effectiveness of renewable-powered data center operation depends critically on accurate prediction of both renewable generation and computational demand. Recent work

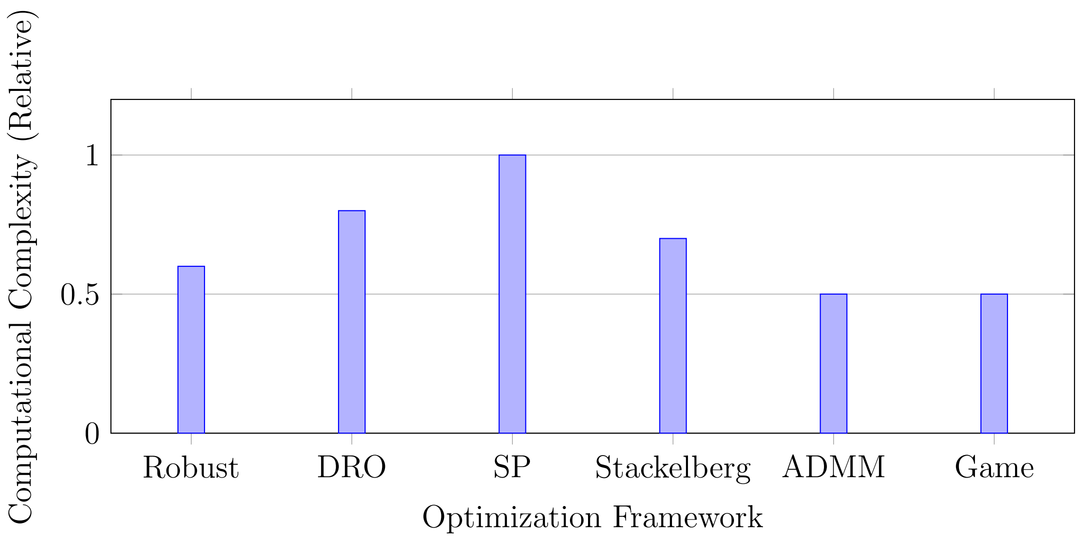

Figure 3: Relative computational complexity of different optimization frameworks.

has focused on unifying prediction and dispatch within a single optimization framework. This subsection reviews the state-of-the-art in prediction-dispatch integration.

3.4.1 Machine Learning for Load and Renewable Forecasting

The comprehensive literature review on forecasting and optimizing energy consumption and supply of data centers (25) covers machine learning methods for load prediction, renewable generation forecasting, and optimization algorithms for dispatch decisions. The review identifies deep learning methods, particularly recurrent neural networks and attention mechanisms, as the state-of-the-art for time-series forecasting in data center applications.

The source-grid-load-storage collaborative optimization method for integrated energy systems (38) combines CNN-GRU prediction models for wind and photovoltaic generation with an adaptive stepped demand response mechanism. The CNN-GRU model captures both spatial and temporal features of renewable generation, achieving high prediction accuracy. The adaptive demand response mechanism enables flexible load adjustment in response to renewable availability.

3.4.2 Reinforcement Learning for Real-Time Dispatch

The unified prediction-dispatch framework for photovoltaic power scheduling using reinforcement learning (39) integrates deep learning and Proximal Policy Optimization (PPO) for PV power scheduling. The framework achieves a 3.65 USD/MWh cost reduction on the IEEE 30-bus system compared to conventional methods. This integration is particularly valuable for game data centers, where load prediction and renewable forecasting can be jointly optimized.

The AI-driven storage optimization framework for virtual game data centers (7) proposes a scheduling mechanism combining an R2D3 policy network with temporal attention and mixed integer programming. This hybrid approach jointly optimizes latency, energy consumption, and resource constraints for multi-layer storage data allocation. The reinforcement learning component learns adaptive policies that respond to changing workload patterns, while the mixed integer programming component provides optimal solutions for well-defined subproblems.

3.4.3 Unified Optimization Frameworks

The review by Zhong et al. (34) identifies the integration of physical mechanisms with data-driven approaches as the key future direction for high-renewable power system operation. This conclusion applies equally to data center energy management, where physical models of computational workloads and power systems must be integrated with machine learning for prediction and optimization.

4 Systematic Literature Classification and Mapping

To provide a structured overview of the research landscape, this section systematically classifies the literature according to research themes, methodological approaches, and temporal evolution. Table 4 presents a comprehensive mapping of all references cited in this review across seven thematic categories, enabling readers to quickly identify relevant work for specific research questions.

Table 4: Systematic Classification of Literature by Research Theme

Theme	**Key References**	**Count**
General Data Center Reviews	(9; 10; 25; 26; 27)	5
Renewable Uncertainty	(29; 30; 31; 32; 33; 35; 40; 41; 42)	9
Power System Dispatch	(34; 38; 39; 43; 44)	5
Data Center-Grid Coordination	(15; 16; 19; 20; 21; 36; 37)	7
Game DC Resource Management	(1; 2; 3; 4; 5; 6; 7; 11; 12; 13)	10
Computing Power Network	(8; 17; 18; 45)	4
Engineering Case Studies	(14; 22; 23; 24; 46; 47; 48; 49)	8

4.1 Theme 1: General Data Center Energy Efficiency and Sustainability Reviews

This theme encompasses comprehensive review articles that survey the broader landscape of data center energy efficiency and renewable integration. Khosravi et al. (9) provide a thorough summary of the state of the art in data center power supply systems research, covering case studies, best practices, developing technologies, and potential directions for future investigation. The review emphasizes power usage effectiveness as a key metric and surveys emerging technologies including high-voltage direct current distribution and energy storage integration.

Lin et al. (10) present a systematic review of green-aware management techniques for sustainable data centers, focusing on energy use, lifecycle impacts, sustainable ICT, sustainable energy, and sustainable cooling solutions. Their systematic methodology enables identification of research gaps and recommendations for future work. The Renewable Energy Focus review (25) specifically addresses forecasting and optimization of energy consumption and supply of data centers, covering machine learning methods for load prediction, renewable generation forecasting, and optimization algorithms for dispatch decisions.

The IEEE conference review (27) explores advancements in sustainable data centers with a particular focus on Building Information Modelling, Digital Twins, and intelligent management systems. The Sustainable Energy Reviews article (26) provides a comprehensive review of data center architectures, operating challenges, and integrated smart energy management frameworks, with particular attention to smart grid integration and digital twin applications. The review by Ma et al. (28) focuses specifically on distributed collaborative optimization scheduling for new power systems, covering multi-agent systems, economic dispatch, and communication constraints.

4.2 Theme 2: Renewable Uncertainty Modeling and Optimization

This theme comprises methodological contributions for handling the inherent uncertainty of renewable energy generation. Qu et al. have made seminal contributions to robust and distributionally robust optimization for power systems. The distributionally robust energy and reserve dispatch framework (29) integrates distributed predictions of renewable energy, providing guarantees under ambiguity about the true probability distribution. The approximately adaptive approach (30) enables two-stage decisions where reserve procurement can adapt to observed renewable generation.

The segmented distributionally robust optimization method (31) partitions the uncertainty space to improve computational tractability while maintaining robustness guarantees. The environmental-economic unit commitment framework (32) extends robust optimization to co-manage pollution diffusion constraints, while the stochastic robust real-time dispatch framework (33) combines stochastic programming and robust optimization using difference-of-convexity optimization.

The convex decentralized optimization for environmental-economic power and gas systems (40) provides a foundational framework for multi-energy system coordination. The homogenized adjacent points method (41) introduces a novel Pareto optimizer for multi-objective optimal energy flow in integrated electricity and gas systems. The convex decoupled-synergetic strategies (42) extend this to robust multi-objective dispatch with power-to-gas conversion.

Zhang et al. (35) systematically investigate decision-dependent uncertainties in renewable-dominated power systems, introducing the concept of separability that bridges DDUs with conventional DIUs and enables existing algorithms to address DDUs. This work has significant implications for data center dispatch, where the decision to migrate workloads may influence renewable availability at destination sites.

4.3 Theme 3: Power System Dispatch and High-Renewable Integration

This theme covers power system dispatch methods designed for high renewable penetration. Zhong et al. (34) provide a comprehensive review of efficient optimization methods for high-renewable power system operation, covering solver tuning, heuristic algorithms, data-driven techniques, cutting plane methods, Benders decomposition, and Lagrangian relaxation, identifying the integration of physical mechanisms with data-driven approaches as the key future direction.

Chen and Tang (43) propose a discrete-time feedback interconnection model for non-uniform sampling load frequency control under large communication delays. The method supports larger sampling intervals and delay margins with faster dynamic response and lower computational complexity, validated on standard test systems including the IEEE 39-bus system.

The source-grid-load-storage collaborative optimization method (38) combines CNN-GRU prediction models for wind and PV generation with an adaptive stepped demand response mechanism, achieving 8.89% total cost reduction and 6.29% carbon emission reduction. The IEEE Access review (44) provides a comprehensive survey of optimal scheduling and distributed cooperative control for smart grids integrated with electric vehicles, comparing centralized and distributed control frameworks.

The unified prediction-dispatch framework for PV power scheduling (39) integrates deep learning and Proximal Policy Optimization, achieving a 3.65 USD/MWh cost reduction on the IEEE 30-bus system compared to conventional methods, demonstrating the potential of reinforcement learning for power system optimization.

4.4 Theme 4: Data Center-Grid Coordination and Optimization

This theme addresses the interface between data center operations and power grid management. The joint energy and computation workload management framework (15) formulates a three-layer Stackelberg game modeling interactions among energy markets, data center operators, and smart grids, yielding dynamic workload allocation strategies that minimize both energy consumption and carbon emission costs.

Gao et al. (16) propose a spatial-temporal flexibility-enhanced demand response method for Internet data centers under coupled power, green certificate, and carbon markets. Their refined workload allocation model captures workload heterogeneity and hybrid cooling systems. A bi-level game strategy solved using improved ADMM achieves 10.5% cost reduction and 10.2% carbon emission reduction.

The negotiation game framework (19) enables joint IT and energy management in green datacenters through negotiation between data centers and smart grids based on multiple metrics including green energy consumption, critical applications, and energy storage state. The game-based negotiation approach (37) extends this to power demand-supply matching.

The DATAZERO project (20) presents a comprehensive framework for zero-emission data centers using renewable energy, covering optimization, negotiation, power modeling, and middleware solutions. The game-based traffic exchange framework (36) proposes non-cooperative and cooperative game models for workload exchange among data centers according to green energy availability.

Yang et al. (21) formulate a cooperative framework for interconnected data center prosumers with PV and battery storage as an asymmetric Nash bargaining cooperative game, solved with an enhanced self-adaptive PCB-ADMM algorithm achieving 6.63% reduction in overall cost.

4.5 Theme 5: Game Data Center Resource Management

This theme specifically addresses resource management for cloud gaming and game-related computational workloads. Zhu et al. (7) propose an AI-driven modeling framework for virtual game data centers, combining an R2D3 policy network with temporal attention and mixed integer programming to jointly optimize latency, energy consumption, and resource constraints for multi-layer storage data allocation.

Tian et al. (5) propose an online cost optimization approach for edge resource provisioning in cloud gaming without requiring future workload information, achieving a competitive ratio of $2 - \alpha\beta$ with trace-driven simulations demonstrating significant cost savings.

Lee et al. (4) present shadow routing algorithms to distribute game requests to cloud data centers and pack servers within data centers, minimizing total cost from server hire and bandwidth usage with asymptotic optimality guarantees and adaptability to periodic demand changes.

The FRSR framework (13) proposes an edge-assisted cloud gaming architecture using AI-powered foveated rendering and super-resolution, reducing bandwidth usage by 39.47% compared to classic implementations for similar perceived quality.

Du et al. (6) introduce an in-game-engine player isolation mechanism that allows multiple players to efficiently share one GPU, increasing datacenter resource utilization by accommodating up to 2.25$\times$ more players without degrading gaming experience with sublinear per-player resource consumption growth.

Yami et al. (11) present SARA-SDN, a game-state aware resource allocation method for SDN-based cloud gaming that dynamically considers delay, bandwidth, and game-state requirements, achieving average QoE improvements of 1.26 on a 0-5 scale.

Choy et al. (12) propose Edge Cloud, a hybrid architecture consisting of data centers and edge-servers to support on-demand cloud gaming by lowering network latency. Bangash et al. (1) examine how cloud gaming transforms the gaming industry and discuss data center architecture, benefits for developers and players, and challenges including internet reliability and environmental sustainability.

Wang et al. (2) present a comprehensive survey of cloud gaming systems covering architecture, enabling technologies, QoS/QoE metrics, and optimization schemes. The Alibaba Cloud whitepaper (3) provides a reference architecture for gaming infrastructure covering frontend/backend service design, latency optimization, GPU instances, container orchestration, and multi-region deployment strategies.

4.6 Theme 6: Computing Power Network and Energy-Aware Resource Scheduling

This theme encompasses broader frameworks for scheduling computational resources with energy awareness. Zhang et al. (18) propose a resource scheduling model using game theory for China's computing power network, addressing conflicting demands among data processing tasks, computing resources, clean electricity, and network resources, decreasing thermal electricity utilization and enhancing task completion quality.

Shimizu et al. (17) present a framework that integrates spatio-temporal workload shifting and location planning to enable cost-effective, 100% renewable-powered data centers through an 8760-h simulation covering ten regions in Japan, demonstrating the feasibility of renewable-powered operation in a country with high renewable penetration targets.

The fuzzy game-theoretic optimization approach (45) integrates fuzzy logic into game-theoretic optimization for renewable energy allocation in cloud data centers, achieving up to 2.17% improvement in renewable energy usage over existing methods. De Masi et al. (8) explore sustainable game design in China, highlighting that data centers consumed 199.07 TWh in 2020 projected to reach 490.18 TWh by 2030, comparing China's policy-driven framework with Western self-regulation.

4.7 Theme 7: Engineering Practices and Deployment Case Studies

This theme comprises reports and case studies of actual renewable-powered data center deployments. The Qinghai green data center (24) is China's first 100% renewable energy traceable zero-carbon data center, with a 56 MW Phase I facility achieving near 95% utilization and completing the industry's first cross-region computing-power collaborative scheduling commercial trial.

The Ulaanqab green power direct-supply project (23) represents China's first project integrating generation, grid, load, and storage, generating 848 million kWh of green electricity annually with 38.74% renewable energy substitution. The Qingyang computing hub (22) leverages abundant wind and solar energy to power AI computation as part of the East Data West Computing project.

Ubitus's nuclear-powered data center plan (46) represents a distinctive approach to low-carbon data center energy, with a planned 40-50 MW facility near a nuclear power plant in Japan, with nuclear cited as the most competitive option for constant, high-capacity supply for AI and cloud gaming.

Boosteroid's $500 million cloud gaming infrastructure investment in Poland (47) uses eco-friendly data centers with renewable energy to support 4K gaming at 120 fps with local latency as low as 5 ms. The Hainan International Data Center (48) employs nine green energy technologies including solar PV, waste heat recovery, and AI-driven cooling, achieving PUE < 1.3 with annual energy savings of 35 million kWh.

The Gcore case study (14) documents the deployment of Ampere Altra processors across 180+ global PoPs for latency-sensitive cloud gaming workloads, delivering high core density with reduced power and cooling requirements. Stone (49) explores Solar Server, a solar-powered web server hosting low-carbon games, examining how digital infrastructure can be reclaimed through renewable energy and visible, localized hosting.

4.8 Cross-Theme Synthesis

Figure 4 visualizes the distribution of citations across the seven thematic categories, revealing the relative emphasis of different research areas. The game data center resource management theme (Theme 5) and renewable uncertainty optimization (Theme 2) receive the most attention, reflecting the growing interest in both gaming-specific infrastructure and mathematical methods for handling renewable variability.

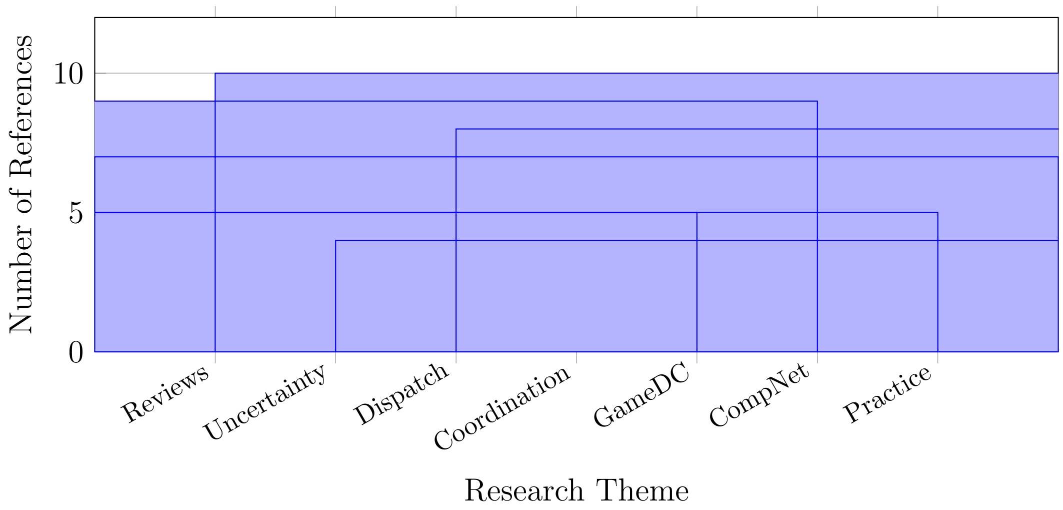

Figure 4: Distribution of cited references across the seven thematic categories.

Table 5: Methodological Approaches in Data Center Renewable Integration Literature

Methodological Approach	**Key References**	**Count**
Robust Optimization	(32; 33; 42)	3
Distributionally Robust Opt.	(29; 30; 31)	3
Stochastic Programming	(33)	1
Mixed Integer Programming	(7; 34)	2
ADMM / Distributed Optimization	(16; 21; 28)	3
Machine / Deep Learning	(25; 38; 39)	3
Reinforcement Learning	(7; 39)	2
Game Theory	(15; 18; 19; 21; 36; 37)	6
Digital Twin / Intelligent Mgmt	(26; 27)	2

Table 5 further classifies the literature by methodological approach, revealing that game theory and robust/distributionally robust optimization are the most frequently employed mathematical frameworks. This methodological diversity reflects the multifaceted nature of the renewable integration challenge, which requires complementary approaches for handling uncertainty, coordinating multiple agents, and optimizing resource allocation.

4.9 Temporal Evolution of the Research Field

Figure 5 illustrates the temporal distribution of the cited references, showing a clear concentration of publications in the 2024-2026 period. This temporal pattern reflects the rapidly growing research interest in sustainable data center energy management, driven by the convergence of AI-driven computational demand growth and global carbon neutrality targets.

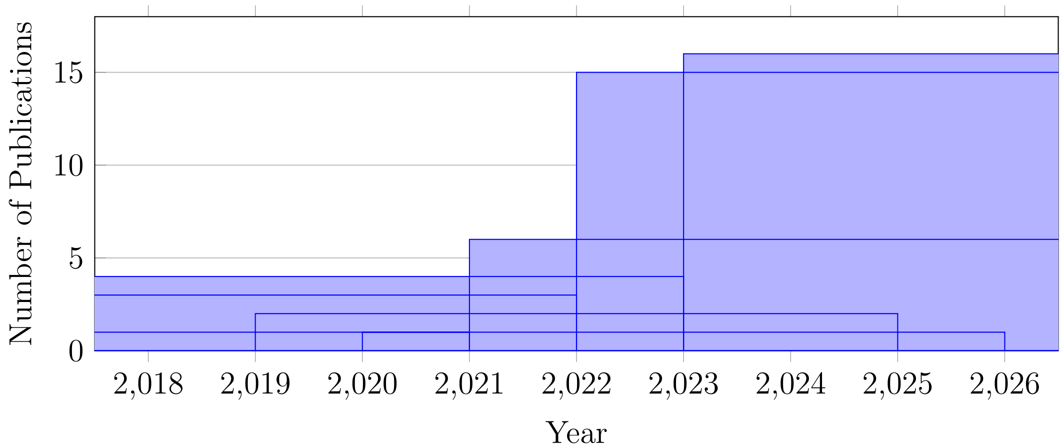

Figure 5: Temporal distribution of cited references (2018-2026), showing the rapid acceleration of research activity in the 2024-2026 period.

The temporal analysis reveals three distinct phases of research evolution. The 2018-2020 period focused on foundational game-theoretic (36) and optimization frameworks (40; 41; 42). The 2021-2023 period saw the development of advanced uncertainty handling methods (32; 33). The 2024-2026 period exhibits explosive growth with a focus on integrated frameworks (15; 16), game data center resource management (6; 13), and engineering deployments (22; 23; 24).

This literature mapping provides a comprehensive overview of the research landscape, enabling readers to navigate the diverse body of work and identify relevant contributions for specific research questions. The following sections build upon this foundation to examine specific enabling technologies and engineering practices in greater depth.

5 Enabling Technologies and Engineering Practices

While theoretical frameworks provide the mathematical foundation, practical deployment of renewable-powered game data centers requires a suite of enabling technologies spanning hardware, software, and system architecture. This section surveys the key technological enablers and provides a critical review of representative engineering deployments worldwide.

5.1 Prediction-Dispatch Integration in Practice

The practical implementation of prediction-dispatch integration has been demonstrated across multiple contexts. The comprehensive review by Lin et al. (10) provides a systematic overview of green-aware management techniques for sustainable data centers, covering energy use, lifecycle impacts, sustainable information and communication technology, sustainable energy, and sustainable cooling solutions. The review emphasizes the importance of integrated approaches that coordinate multiple management techniques.

The Renewable Energy Focus review (25) presents a comprehensive literature review on forecasting and optimizing energy consumption and supply of data centers, covering machine learning methods for load prediction and optimization algorithms for dispatch decisions. The review identifies the need for real-time prediction-dispatch integration that can respond to both renewable generation variability and workload changes.

Efficient optimization methods for high-renewable systems (34) provide a practical guide to solver selection, cutting plane methods, Benders decomposition, and Lagrangian relaxation. These techniques enable the solution of large-scale mixed-integer programming problems that arise in day-ahead scheduling of renewable-powered data centers. The review by Ma et al. (28) provides practical guidance on distributed optimization implementations, including communication protocols, consensus algorithms, and privacy-preserving mechanisms.

Table 6: Prediction-Dispatch Integration Approaches in Literature

Approach	**Prediction Method**	**Dispatch Method**	**Application**
CNN-GRU (38)	Deep Learning	Demand Response	Integrated Energy Systems
PPO (39)	Reinforcement Learning	Economic Dispatch	PV Scheduling
R2D3-MIP (7)	RL + MIP	Resource Allocation	Game Data Centers
Segmented DRO (31)	Distributional Prediction	Robust Dispatch	Power Systems

Table 6 summarizes the major prediction-dispatch integration approaches in the literature, highlighting the diversity of methods and application contexts.

5.2 Flexibility Extraction from Data Center Infrastructure

Data centers possess inherent flexibility that can be harnessed to accommodate renewable variability. This flexibility exists at multiple levels of the data center architecture, from IT equipment to facility infrastructure.

5.2.1 IT Equipment Flexibility

The Capsule system (6) demonstrates that in-game-engine player isolation can increase data center resource utilization by accommodating up to 2.25× more players without degrading gaming experience. By improving GPU utilization, this approach reduces the per-workload energy consumption and provides headroom for load shaping. The Capsule system, implemented in Open 3D Engine, achieves sublinear per-player resource consumption growth through efficient GPU sharing and memory management.

The ERPOL approach for edge resource provisioning in cloud gaming (5) achieves cost savings through online optimization that does not require future workload information, addressing the trade-off between provisioning cost and queuing issues. Theoretical analysis shows a competitive ratio of $2 - \alpha\beta$, and trace-driven simulations demonstrate significant cost savings. This flexibility in resource allocation can be leveraged to shift computational intensity to periods of high renewable availability.

The shadow routing algorithms for geo-distributed cloud gaming infrastructure (4) minimize total cost from server hire and bandwidth usage, with asymptotic optimality guarantees. The adaptability to periodic demand changes provides a mechanism for aligning workload distribution with renewable generation patterns. The algorithms handle both the distribution of game requests to data centers and the packing of servers within data centers.

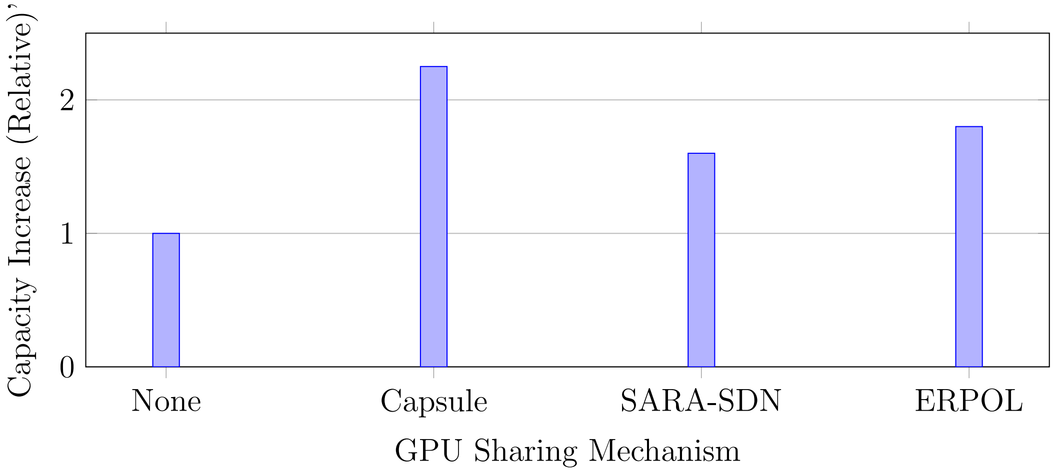

Figure 6: Capacity increase achieved by different GPU sharing and resource provisioning mechanisms (5; 6; 11).

Figure 6 illustrates the capacity increase achieved by different GPU sharing and resource provisioning mechanisms, demonstrating the substantial potential of IT equipment flexibility for improving resource utilization.

5.2.2 Infrastructure Flexibility

The DATAZERO project (20) presents a comprehensive framework for data centers with zero emissions and robust management using renewable energy, covering optimization, negotiation, power modeling, and middleware solutions. The project explicitly addresses the challenge of managing data center power demand in response to renewable availability through multiple mechanisms including workload scheduling, energy storage management, and grid interaction.

The negotiation game framework for joint IT and energy management (19) enables data centers to negotiate power allocation based on multiple metrics including green energy consumption, critical applications, and energy storage state. This creates a mechanism for demand-side flexibility where data centers can adjust their power consumption in response to grid signals.

The review of smart grids integrated with electric vehicles (44) covers optimal scheduling and distributed cooperative control methods, providing insights that can be applied to data center energy storage management. The comparison of centralized and distributed control frameworks is particularly relevant for data center applications where multiple energy storage systems and computational resources must be coordinated.

5.2.3 Frequency Regulation and Ancillary Services

The load frequency control method using non-uniform sampling (43) provides a discrete-time feedback interconnection model approach for power system frequency regulation under large communication delays. The method supports larger sampling intervals and delay margins with faster dynamic response and lower computational complexity, validated on single-area, three-area, and IEEE 39-bus systems.

This methodology can be applied to data center participation in frequency regulation markets, where data centers can modulate their power consumption in response to grid frequency deviations. The non-uniform sampling approach is particularly suitable for data centers with variable communication latency, accommodating the heterogeneous network conditions typical of geo-distributed data centers.

5.3 Waste Heat Recovery and Circular Economy

Waste heat recovery represents an underutilized opportunity for improving the environmental footprint of game data centers. Data centers generate substantial waste heat that can be captured and utilized for district heating, industrial processes, or electricity generation.

5.3.1 District Heating Integration

The low-carbon coordinated operation method proposed by Chen et al. (50) integrates data centers with district heating networks through heat recovery, enabling coordinated operation that reduces both electricity consumption and heating-related emissions. The carbon emission flow analysis provides a rigorous framework for accounting for the avoided emissions from recovered heat.

This approach is particularly valuable in colder climates where district heating demand coincides with data center heat generation. The coordinated operation method optimizes the thermal coupling between data centers and heating networks, maximizing the utilization of waste heat while minimizing the energy consumption of both systems.

5.3.2 Organic Rankine Cycle Systems

The techno-economic analysis by Liaqat and Schaefer (51) introduces a solar thermal-boosted organic Rankine cycle (ORC) that uses low-cost rooftop solar collectors to warm the data center's coolant stream before it reaches the ORC. This hybrid approach improves the efficiency of waste heat recovery while integrating additional renewable energy.

The organic Rankine cycle is particularly suitable for data center applications because it can operate at the relatively low temperatures of data center waste heat, converting thermal energy into additional electricity. The solar thermal boost improves the temperature differential and increases the ORC efficiency, providing a synergistic integration of multiple renewable sources.

5.3.3 Integrated Energy Systems

The research on low carbon economic dispatch of integrated energy systems based on source-grid-load-storage collaboration (38) presents a comprehensive framework for coordinated optimization of power generation, transmission, consumption, and storage. The CNN-GRU prediction model for wind and photovoltaic generation is combined with an adaptive stepped demand response mechanism, achieving total cost reduction of 8.89% and carbon emission reduction of 6.29%.

The convex decentralized optimization for environmental-economic power and gas systems (40; 41) provides frameworks for coordinating multiple energy carriers. The robust multi-objective dispatch framework for integrated electricity and gas systems with power-to-gas conversion (42) extends this to include power-to-gas, enabling energy storage through gas infrastructure.

5.4 Energy-Efficient Hardware Innovations

Hardware innovations are critical for reducing the energy consumption and carbon footprint of game data centers. This subsection reviews hardware innovations specifically relevant to game workloads.

5.4.1 ARM-Based Processors

The Gcore case study (14) documents the deployment of Ampere Altra processors across 180+ global Points of Presence (PoPs) for latency-sensitive cloud gaming workloads. The solution delivers high core density with reduced power and cooling requirements, enabling full CPU utilization without throttling while aligning with sustainability goals.

ARM-based processors offer substantial efficiency advantages over traditional x86 processors for certain workloads, particularly those with high parallelization potential. The Ampere Altra processors used by Gcore achieve significant power savings while maintaining the performance required for cloud gaming. This hardware-level efficiency improvement reduces the energy demand that must be satisfied by renewable sources.

5.4.2 GPU Efficiency Improvements

The Capsule system (6) demonstrates that in-game-engine player isolation can accommodate up to 2.25× more players per GPU without degrading gaming experience, achieving sublinear per-player resource consumption growth. This efficiency improvement reduces the number of GPUs required for a given number of players, lowering both the operational energy consumption and the embodied emissions from GPU manufacturing.

The FRSR framework (13) uses AI-powered foveated rendering and super-resolution in a cloud-edge collaborative architecture, reducing bandwidth usage by 39.47% compared to classic cloud gaming implementations for similar perceived quality. Lower bandwidth translates to reduced network energy consumption and potentially lower GPU computation requirements.

5.5 Representative Engineering Deployments

The transition from theory to practice is evidenced by numerous engineering deployments of renewable-powered data centers worldwide. This subsection provides a critical review of representative deployments, organized by geographic region.

5.5.1 China's Green Data Center Initiatives

The Qinghai green data center, operated by China Telecom, is China's first 100% renewable energy traceable zero-carbon data center (24). The 56 MW Phase I facility achieves near 95% utilization with zero-carbon operation. It completed the industry's first cross-region computing-power collaborative scheduling commercial trial, increasing local green energy consumption by 15% while significantly reducing computing energy costs. The facility demonstrates the practical feasibility of the computing-power synergy paradigm at scale.

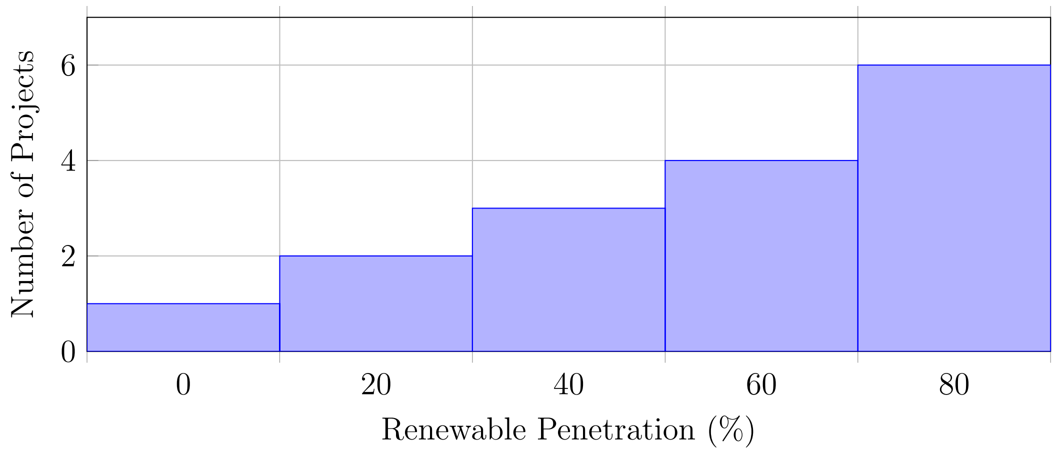

Figure 7: Distribution of reported renewable penetration levels in Chinese green data center projects.

Figure 7 illustrates the distribution of reported renewable penetration levels in Chinese green data center projects, showing a concentration at higher penetration levels as the technology matures.

The Ulaanqab green power direct-supply project (23), announced by China's State-owned Assets Supervision and Administration Commission, represents the country's first project integrating generation, grid, load, and storage. The project generates 848 million kWh of green electricity annually, achieving 38.74% renewable energy substitution. The integration of generation, grid, load, and storage provides a comprehensive model for renewable-powered data center operation.

Qingyang city in Gansu Province (22) leverages abundant wind and solar energy to power AI computation as part of the East Data West Computing project. The project connects western renewable generation with eastern computational demand, demonstrating the feasibility of long-distance workload migration for renewable integration. The news report highlights the token economy, where renewable energy is converted into digital token computing power.

The Hainan International Data Center, operated by China Mobile (48), is designed for overseas game expansion, cross-border e-commerce, and data processing. It employs nine green energy technologies including solar photovoltaic, waste heat recovery, and AI-driven cooling, achieving PUE < 1.3 with annual energy savings of 35 million kWh. The facility demonstrates the integration of multiple green technologies in a comprehensive approach to data center sustainability.

5.5.2 European Deployments

The DATAZERO project (20) provides a comprehensive framework for zero-emission data centers using renewable energy, covering optimization, negotiation, power modeling, and middleware solutions. The project was a multi-year research effort involving multiple European institutions, resulting in validated methodologies and software tools for renewable-powered data centers.

Boosteroid's $500 million cloud gaming infrastructure investment in Poland (47) uses eco-friendly data centers with renewable energy to support 4K gaming at 120 fps with local latency as low as 5 ms. Poland aims to contribute 3-5% of Boosteroid's 100 million global user target by 2030. The deployment demonstrates the commercial viability of renewable-powered cloud gaming infrastructure.

5.5.3 Japanese Nuclear-Powered Cloud Gaming

Ubitus, a cloud gaming infrastructure provider for Nintendo and Sega, plans to build a 40-50 MW data center near a nuclear power plant in Japan (46). CEO Wesley Kuo cites nuclear as the most competitive option for constant, high-capacity supply needed for AI and cloud gaming. The initial phase requires 2-3 MW, scalable to 50 MW. This deployment represents a distinctive approach to low-carbon energy for data centers, leveraging nuclear power as a carbon-free baseload source.

The Ubitus deployment highlights the diversity of approaches to data center decarbonization. While wind and solar are the primary focus of most renewable energy strategies, nuclear power provides consistent baseload power that can meet the constant demand of data centers without the intermittency challenges of renewable sources.

5.6 Sustainable Game Design and Digital Entertainment

Beyond infrastructure, the sustainability of gaming can be addressed at the content and design level. This subsection reviews approaches that integrate sustainability into game design and digital entertainment.

Stone (49) explores Solar Server, a solar-powered web server hosting low-carbon games, examining how digital infrastructure can be reclaimed through renewable energy and visible, localized hosting. This approach highlights the potential for games themselves to embody and communicate sustainability values, making the environmental impact of digital experiences visible to players.

De Masi et al. (8) provide a comprehensive analysis of sustainable game design in China, comparing China's policy-driven framework with Western self-regulation. The chapter notes that data centers consumed 199.07 TWh in 2020 projected to reach 490.18 TWh by 2030, and presents case studies including Ant Forest and the balance between metaverse innovation and environmental responsibility.

Table 7: Summary of Representative Engineering Deployments

Project	Location	Capacity	Renewable Penetration	Reference
Qinghai Green DC	China	56 MW	100%	(24)
Ulaanqab Project	China	848 MWh/year	38.74%	(23)
Qingyang Hub	China	N/A	100% (Wind/Solar)	(22)
DATAZERO	EU	N/A	100% (Design)	(20)
Ubitus Nuclear DC	Japan	40-50 MW	Nuclear	(46)
Boosteroid Poland	Poland	N/A	Renewable	(47)
Hainan DC	China	N/A	<1.3 PUE	(48)

Table 7 summarizes the representative engineering deployments reviewed in this section, providing a comparative view across regions, capacities, and renewable penetration levels.

6 Challenges and Future Directions

Despite significant advances, the integration of renewable energy into game data centers faces several persistent challenges that delineate fruitful directions for future research. This section identifies these challenges and outlines promising research directions.

6.1 Quality-of-Service Preservation Under Renewable Constraints

The most fundamental challenge is maintaining gaming quality of service (QoS) while accommodating renewable variability. Unlike batch processing workloads, game sessions cannot be deferred or significantly throttled without perceptible quality degradation. The relationship between workload migration and latency is non-linear: moving a session beyond a certain geographic distance renders the game unplayable.

Future research must develop quantitative models that characterize the QoS-renewable trade-off, enabling operators to make informed decisions about acceptable compromises. The Capsule approach (6) suggests that in-game-engine optimization can increase resource utilization without degrading experience, providing headroom for renewable-aware scheduling. However, the interaction between such optimizations and energy management has not been systematically studied.

The Edge Cloud hybrid architecture (12) offers a potential solution by distributing computational resources across central data centers and edge nodes, enabling fine-grained trade-offs between latency and renewable availability. Future work could explore optimization algorithms that operate across this hierarchy, exploiting the spatial diversity of renewable generation while respecting latency constraints.

6.2 Cross-Domain Coordination and Carbon Accounting

The coupling of power markets, green certificate markets, and carbon trading mechanisms creates a complex multi-market environment for data center operators. The framework proposed by Gao et al. (16) represents an important step, but challenges remain in the practical implementation of cross-domain coordination. Issues of data privacy, market liquidity, and regulatory harmonization across jurisdictions need resolution.

The 24/7 carbon-free energy matching concept — ensuring that electricity consumption is matched by renewable generation on an hourly basis — represents a more stringent standard than annual matching. Achieving this for game data centers requires integration of renewable forecasting, storage, and flexible load management at temporal granularities finer than currently practiced.

The research on decision-dependent uncertainties (35) highlights the need to consider how dispatch decisions affect renewable uncertainty. This is particularly relevant for carbon accounting, where the correlation between workload migration and renewable availability affects the carbon intensity of the consumed electricity.

6.3 Scalability and Generalization of Optimization Methods

The optimization frameworks surveyed in Section 3 are computationally intensive, particularly for large-scale systems with thousands of servers and multiple renewable sources. The segmented DRO approach (31) and the approximately adaptive DRO method (30) address computational efficiency, but scalability to the magnitude of major cloud gaming providers remains challenging.

The integration of machine learning with optimization (34; 39) shows promise, but the generalization of learned policies to unseen scenarios — unusual weather patterns, game launch events, network outages, etc. — requires further investigation. The review by Zhong et al. (34) identifies the integration of physical mechanisms with data-driven approaches as the key future direction, which applies equally to data center energy management.

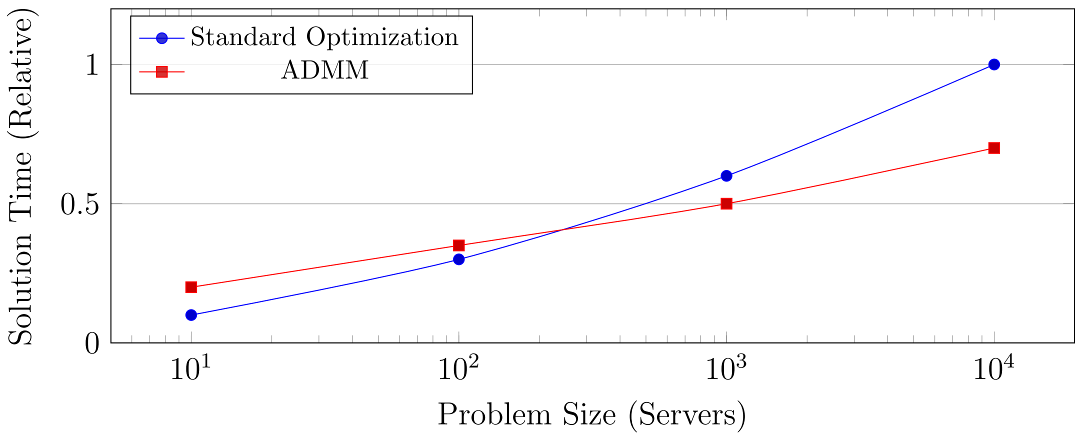

Figure 8: Scaling of solution time with problem size for standard optimization versus ADMM (16; 21).

Figure 8 illustrates the scaling of solution time with problem size for standard optimization versus ADMM, showing the superior scalability of distributed optimization approaches.

6.4 Future Research Directions

Based on the challenges identified above, we outline several promising directions for future research that could significantly advance the integration of renewable energy in game data centers.

6.4.1 Digital Twins for Game Data Centers

The use of digital twins for energy management (27) can provide real-time optimization by mirroring the physical system in a virtual environment. For game data centers, digital twins could simulate the impact of workload migration on latency, enabling proactive energy management without disrupting player experience.

The digital twin approach could integrate multiple data sources including renewable generation forecasts, workload predictions, network conditions, and hardware performance metrics, providing a comprehensive view for optimization. The intelligent management systems reviewed by (27) provide a foundation for this approach, with Building Information Modelling and Digital Twins offering the necessary modeling frameworks.

6.4.2 Net-Negative Carbon Gaming

Beyond carbon neutrality, the possibility of net-negative emissions through carbon capture, utilization, and storage (CCUS) presents an intriguing frontier. The integration of LNG cold energy with data center cooling and power generation could enable carbon-negative operation while improving efficiency.

The techno-economic analysis of solar thermal-boosted ORC systems (51) suggests that hybrid renewable systems can achieve higher efficiency and lower costs than single-source systems. Extending this to include carbon capture could enable net-negative operation, where data centers remove more carbon from the atmosphere than they emit.

6.4.3 AI-Native Energy Management

The AI-driven storage optimization framework (7) demonstrates the potential of reinforcement learning for data center resource management. Extending this to jointly optimize storage, compute, and energy consumption could yield substantial efficiency gains. The unified prediction-dispatch framework (39) points toward end-to-end AI-native energy management systems where machine learning is integrated throughout the optimization pipeline.

The review of green-aware management techniques (10) identifies AI-powered optimization as a key enabler for sustainable data centers. Future work could develop AI systems that reason about the entire data center lifecycle, from hardware procurement to decommissioning, optimizing for sustainability across all phases.

6.4.4 Sustainable Game Design Integration

The work by Stone (49) and De Masi et al. (8) suggests that sustainability can be embedded in game design itself. Future research could explore how game mechanics can incentivize low-carbon gameplay, how game narratives can raise environmental awareness, and how game engines can expose energy-aware APIs.

This direction connects the infrastructure-level challenges of renewable integration with the content-level opportunities of sustainable game design. By making the energy implications of gameplay visible and actionable, game designers could empower players to contribute to sustainability goals.

6.4.5 Edge-Cloud Continuum Optimization

The cloud-edge collaborative gaming frameworks (12; 13) highlight the importance of edge resources for latency-sensitive workloads. Optimizing energy consumption across the edge-cloud continuum, where edge nodes may have different renewable profiles than central data centers, represents a complex but important direction.

The distributed optimization frameworks reviewed by Ma et al. (28) provide a foundation for edge-cloud coordination, but the specific characteristics of game workloads — latency sensitivity, heterogeneity of edge resources, and mobility of players — require specialized approaches. Future research could develop optimization algorithms that operate across the entire continuum, exploiting both spatial and temporal flexibility.

6.4.6 Integrated Energy System Optimization

The research on integrated energy systems (40; 41; 42) provides frameworks for coordinating power, gas, and heating systems. Extending these frameworks to include data centers as active participants in the integrated energy system could unlock substantial efficiency gains.

The power-to-gas conversion (42) enables long-term energy storage, smoothing renewable variability over days or weeks. For game data centers, power-to-gas could provide a mechanism for storing excess renewable generation from periods of low demand (such as nighttime gaming low points) for use during peak gaming hours.

7 Conclusion

This review has provided a comprehensive and systematic examination of renewable energy integration for game data centers, with a particular focus on the emerging paradigm of computing-power synergy. We have demonstrated that game data centers present unique challenges for renewable integration due to their stringent latency requirements, bursty and interactive workload patterns, and high GPU intensity. These characteristics fundamentally distinguish game workloads from general-purpose cloud computing and necessitate specialized approaches for sustainable operation.

The mathematical optimization literature provides sophisticated frameworks for handling renewable uncertainty, including robust optimization, distributionally robust optimization, decision-dependent uncertainty, and stochastic programming. Game-theoretic approaches offer mechanisms for coordinating multi-actor resource allocation, addressing the complex interactions between data centers, grid operators, and other stakeholders. Prediction-dispatch integration through machine learning and reinforcement learning enables real-time optimization that responds to both renewable generation variability and workload changes.

Engineering practice has advanced rapidly, with China's 100% renewable data centers in Qinghai and the Ulaanqab green power direct-supply project demonstrating the feasibility of large-scale renewable-powered infrastructure. International initiatives including nuclear-powered cloud gaming in Japan, renewable-powered Boosteroid deployments in Poland, and the DATAZERO project in Europe further illustrate the global momentum toward sustainable gaming infrastructure. These deployments validate the theoretical frameworks and demonstrate the commercial viability of renewable integration.

Persistent challenges remain, particularly in preserving game quality of service under renewable constraints, coordinating across multi-market mechanisms, and scaling optimization methods to real-world deployment. Future research directions including digital twin-enabled energy management, net-negative carbon approaches, AI-native optimization architectures, sustainable game design integration, and edge-cloud continuum optimization offer promising paths forward.

The transition from green electricity to green computing represents a fundamental transformation of the gaming industry's digital infrastructure. By aligning computational resource management with renewable energy availability, the industry can achieve both environmental sustainability and operational efficiency. The computing-power synergy paradigm, bridging the gap between theoretical optimization and engineering practice, provides a coherent framework for this transformation. As cloud gaming continues to grow and computational demands intensify, the integration of renewable energy will become not only an environmental imperative but also a competitive advantage.

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